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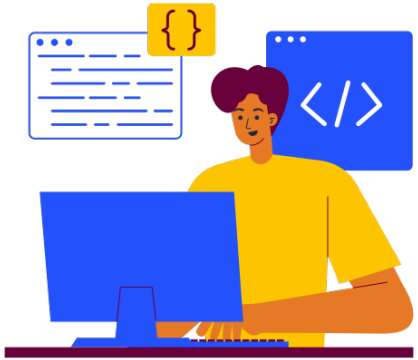
SINGAPORE INSTITUTE OF TECHNOLOGY

Full Stack Secured Development

Sequelize ORM and Input Validation



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Learning Outcomes

By the end of this lesson, you will be able to:

1. Describe Sequelize ORM.
2. Understand Sequelize CRUD Operations.
3. Describe Sequelize Associations.
4. Implement Input Validation using Yup.

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Sequelize ORM

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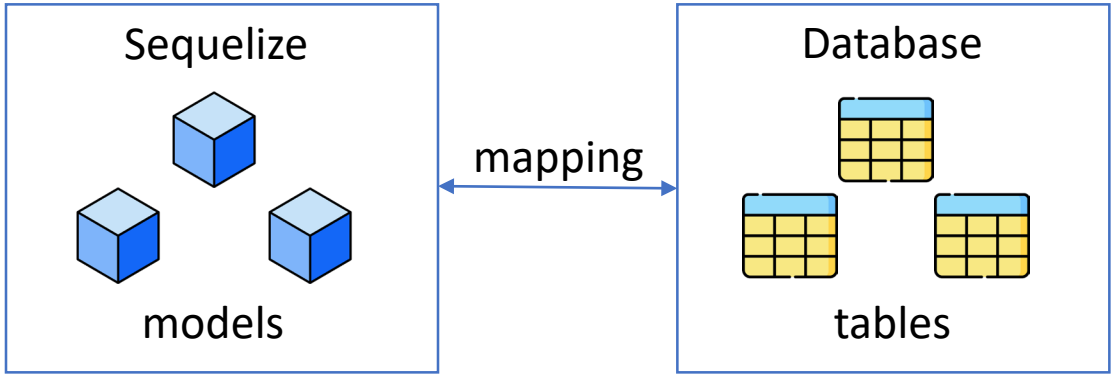
Sequelize ORM

Sequelize v6 used

- **Sequelize** is an Object-Relational Mapping (ORM) library for Node.js that provides a layer between your application and the database.
- It allows you to work with objects and perform database operations using JavaScript methods, rather than writing raw SQL queries.



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Key Features

Here are some key features of Sequelize:

- **Database Agnostic**: supports multiple relational databases, including MySQL, PostgreSQL and SQLite
- **Model-Driven**: allows you to define models that represent database tables
- **CRUD Operations**: provides intuitive methods for creating, reading, updating, and deleting records in the database
- **Association Management**: allows you to define relationships between models using associations, such as one-to-many relationship

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Installing

Open terminal or command prompt

```
> npm i sequelize
```

- Install sequelize

Install the driver for your database of choice:

```
> npm i mysql2          # MySQL
```

- Known for its speed, reliability, and ease of use

```
> npm i pg pg-hstore    # PostgreSQL
```

- Known for its advanced features and extensibility

```
> npm i sqlite3         # SQLite
```

- Lightweight local database

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Sequelize Instance

To connect to the database, you must create a Sequelize instance.

```
// Create sequelize instance for MySQL
let sequelize = new Sequelize(
  process.env.DB_NAME,
  process.env.DB_USER,
  process.env.DB_PWD,
  {
      host: process.env.DB_HOST,
      port: process.env.DB_PORT,
      dialect: 'mysql',
      logging: false,
      timezone: '+08:00'
  }
);
```

- The code uses process.env to access **environment variables** for sensitive configuration like the database name, username, password, host, and port.
- **logging: false**: This option disables logging of SQL queries to the console. If set to true, Sequelize would log all executed queries.
- **timezone: '+08:00'**: This option sets the timezone for the database connection. In this case, it's set to UTC+08:00 for Singapore local time.

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Sequelize Model

Sequelize models are used for defining the structure of the database table, specifying its columns and data types, and defining associations with other Sequelize models.

```
module.exports = (sequelize, DataTypes) => {
    const Product = sequelize.define("Product", {
        name: {
            type: DataTypes.STRING,
            allowNull: false,
        },
        price: {
            type: DataTypes.DECIMAL(10, 2),
            allowNull: false,
        }
    }, {
        tableName: 'products'
    });

    return Product;
}
```

- Model name: **Product**
- Attribute name is using **string** data type
- Attribute price is using **decimal** number with 10 digits and 2 decimal places
- Table name: **products**
- Sequelize will automatically generate the **id** field as the primary key, with integer type.

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Data Types

Commonly used data types in Sequelize:

- **STRING**: Represents a variable-length string. You can specify the maximum length using STRING(length), e.g. STRING(50).
- **TEXT**: Represents a longer text without a specific length limit.
- **INTEGER**: Represents an integer number.
- **DECIMAL**: Represents a decimal number. You can specify the total number of digits and the number of decimal places, e.g. DECIMAL(10, 2) for price.
- **BOOLEAN**: Represents a boolean value (true or false).
- **DATE**: Represents a date with time.
- **DATEONLY**: Represents a date without time.

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Synchronizing Models

sync is a Sequelize method that synchronizes the models with the database schema.

```
sequelize.sync({ alter: true })
    .then(() => {
        // start your server here
    })
    .catch((err) => {
        console.log(err);
    });
```

- Setting **alter** to **true** tells Sequelize to make any necessary changes to the database schema to match the model definitions.

- **Synchronize** is commonly used during the project **development** to automatically make any necessary changes to the database structure without requiring manual intervention.
- However, for **production**, it's recommended to use database **migrations** for more controlled schema changes, to prevent data loss or unexpected consequences.

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CRUD Operations

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Create

endpoint ‘POST /product’: create a new product

routes/product.js

```
const { Product } = require('../models');

// Define a POST endpoint to create a new product
router.post("/", async (req, res) => {
    let data = req.body;
    let result = await Product.create(data);
    res.json(result);
});
```

- Import sequelize model
- Use the Product model **create()** method to insert one row into the database table using the data received in the request body

Asynchronous Functions (with await)

- Sequelize provides **asynchronous** methods that return Promises. Using **await** with these methods allows you to write non-blocking code, which is the preferred approach in Node.js applications.

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Read (find all)

endpoint ‘GET /product’: list all products

routes/product.js

```
const { Op } = require('sequelize');

// Define a GET endpoint to list all products
router.get('/', async (req, res) => {
    let minPrice = parseFloat(req.query.minPrice) || 0;
    let list = await Product.findAll({
        // Options for filtering and ordering results
        where: {
            price: { [Op.gte]: minPrice }
        },
        order: [['price', 'ASC']],
    });
    res.json(list);
});
```

- Read minimum price from query parameters
- The method **findAll()** returns an array of model instances, where each instance corresponds to a row in the database table.
- Option **where**: Filter products with a price greater than or equal to the minimum value
- Option **order**: Order the results by the 'price' column in ascending order

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Sequelize Operators

Sequelize provides a range of operators that you can use to define conditions in your database queries. They are available in the **Op** object.

Implicit Operators

```
brand: null
```

- **Op.is**: brand is null

```
brand: 'Apple'
```

- **Op.eq**: brand equals ‘Apple’

```
brand: ['Apple', 'Samsung']
```

- **Op.in**: brand in the array with ‘Apple’, ‘Samsung’

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Sequelize Operators

Comparison Operators

- Op.gt: Greater than
- Op.gte: Greater than or equal to
- Op.lt: Less than
- Op.lte: Less than or equal to

```
price: { [Op.gte]: 50 }
```

Between Operator

- Op.between: Between two values

```
price: { [Op.between]: [50, 100] }
```

Like Operator

- Op.like: Matches a string pattern

```
name: { [Op.like]: '%iPhone%' }
```

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Logical Operators

AND

- A single object can specify multiple conditions to check.
- If you do not specify any combination operator, Sequelize will default to joining them with **Op.and** operator.

```
{
    category: 'Electronics',
    price: { [Op.gt]: 100 }
}
```

- Implicit **Op.and**: category equals ‘Electronics’ **and** price greater than 100

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Logical Operators

OR

- To replace that implicit AND with an OR, you can use the **Op.or** operator.

```
[Op.or]: [
    { category: 'Electronics' },
    { price: { [Op.gt]: 100 } }
]
```

- **Op.or**: category equals ‘Electronics’ **or** price greater than 100

```
[Op.or]: [
    {
        category: 'Electronics',
        active: true
    },
    { price: { [Op.gt]: 100 } }
]
```

- OR operator only applies to one level of nesting at a time
- Condition: (category equals ‘Electronics’ **and** active is true) **or** price greater than 100

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Logical Operators

NOT

- The operator **Op.not** is used to negate the condition.

```
category: { [Op.not]: 'Clothing' }
```

- **Op.not**: category is not ‘Clothing’

```
[Op.not]: {
    category: 'Electronics',
    price: { [Op.gt]: 100 }
}
```

- **Op.not**: not (category equals ‘Electronics’ and price greater than 100)

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Read (find by PK)

endpoint ‘GET /product/{id}’: retrieve a product by ID

routes/product.js

```
// Define a GET endpoint to get a product by ID
router.get('/:id', async (req, res) => {
    const id = req.params.id;
    const product = await Product.findByPk(id);
    if (!product) {
        res.sendStatus(404);
        return;
    }
    res.json(product);
});
```

- The method **findByPk()** is used to retrieve a single record from a table based on its primary key value.

- Sequelize also provides another method **findOne()** to retrieve a single record with where condition.

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Update

endpoint ‘PUT /product/{id}’: update a product by ID

routes/product.js

```
// Define a PUT endpoint to update a product by ID
router.put('/:id', (req, res) => {
    const id = req.params.id;
    const product = await Product.findByPk(id);
    if (!product) {
        res.sendStatus(404);
        return;
    }
    // Update the product
    let data = req.body;
    await Product.update(data, { where: { id: id } });
    res.json({ message: 'Product updated' });
});
```

- The method **update()** with the where condition will find the record with the provided id and update the record with the new data.

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Delete

endpoint ‘DELETE /product/{id}’: delete a product by ID

routes/product.js

```
// Define a DELETE endpoint to delete a product by ID
router.delete('/:id', (req, res) => {
    const id = req.params.id;
    const product = await Product.findByPk(id);
    if (!product) {
        res.sendStatus(404);
        return;
    }
    // Delete the product
    await Product.destroy({ where: { id: id } });
    res.json({ message: 'Product deleted' });
});
```

- The method **destroy()** with the where condition will delete the record with the provided id.

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Associations

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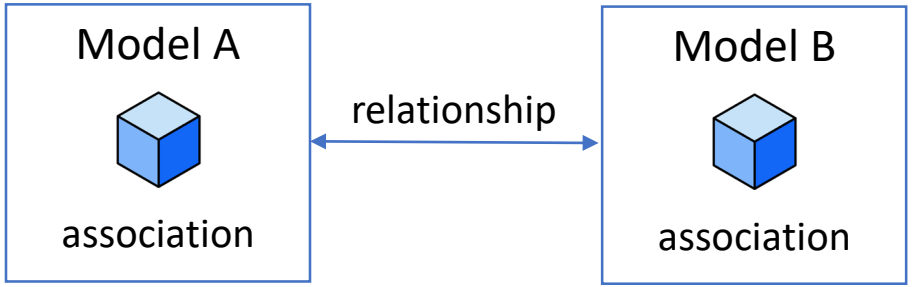
Sequelize Associations

Sequelize **associations** are used to define **relationships** between tables (models) in a relational database.

- One-to-One relationship: use hasOne and belongsTo associations
- One-to-Many relationship: use hasMany and belongsTo associations
- Many-to-Many relationship: use two belongsToMany associations



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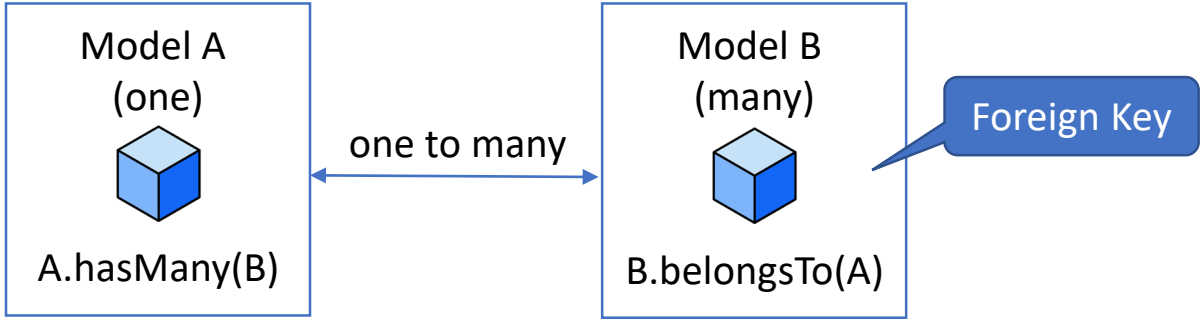
One-to-Many Relationship

- The **one-to-many** relationship is a common type of database relationship in which one record in one table is associated with multiple records in another table.
- In Sequelize, you can define and work with one-to-many relationships using associations **hasMany** and **belongsTo**.
- One-to-Many relationship example:
  - One Category (Model A) has many Products (Model B)



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hasMany Association

- One Category has many Products

models/Catogory.js

```
Category.associate = (models) => {
    Category.hasMany(models.Product, {
        onDelete: "cascade"
    });
};
```

- The method **hasMany()** defines the association for the one-to-many relationship at the one side.
- **onDelete: “cascade”** means if a category is deleted, all its associated products are deleted as well.

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belongsTo Association

- One Product belongs to one Category

models/Product.js

```
Product.associate = (models) => {
    Product.belongsTo(models.Category, {
        foreignKey: "categoryId",
        as: "category"
    });
};
```

- The foreign key will be generated automatically at the many side (Product).
- The default foreign key column name in the Product table will be categoryId.

- The method **belongsTo()** defines the association for the one-to-many relationship at the many side.
- To use your custom foreign key name, you can specify it explicitly using the **foreignKey** option.
- The option **as** defines the alias, which allows you to access the associated Category model using a preferred name when querying.

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include Option

- Include Category when querying Product

```
// Define a GET endpoint to list all products
router.get('/', async (req, res) => {
    let minPrice = parseFloat(req.query.minPrice) || 0;
    let list = await Product.findAll({
        where: {
            price: { [Op.gte]: minPrice }
        },
        include: {
            model: Category,
            as: "category",
            attributes: ['name']
        },
        order: [['price', 'ASC']],
    });
    res.json(list);
});
```

- The **include** option is used to include related models when querying the main model. It enables eager loading of associated data in a single query.
- **model: Category** specifies the model to include
- **as: "category"** specifies the alias for the included model, same as the alias defined in the belongsTo association
- **attributes: ['name']** specifies which columns from the included model you want to retrieve
- The include option can also be used in other query methods like findByPk.

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Input Validation

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Input Validation

Web API input validation refers to the process of validating the data sent to a web API, especially in the request body.

- **Data Type** Validation: matches expected data type, e.g. string, number
- **Length** Validation: check minimum and maximum length for string
- **Range** Validation: check minimum and maximum range for number or date
- **Format** Validation: matches expected format for string, e.g. email, url, regular expression
- **Whitelist** Validation: only allows value in a whitelist

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Popular Libraries

- Express-validator
  - If you are building a Node.js web API using Express.js and need server-side validation and sanitization, Express-validator is a great choice.
- Yup
  - If you are working on a front-end application (e.g., a React app) and need client-side validation with reusable schemas that can be shared between the client and server, Yup is a strong option.

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Yup

Yup is a general-purpose JavaScript validation library suitable for both front-end and back-end.

- **Schema Definition**: You can define a validation schema that describes the structure and validation rules for the input data.
- **Chainable Methods**: You can chain validation functions together to create complex validation rules.
- **Data Shaping**: Yup not only validates data but can also transform it based on your schema.
- **Error Messages**: You can specify the custom error message for each validation rule.

```
npm i yup
```

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Define Schema

- Define validation schema when creating product

```
let data = req.body;
// Validate request body
let validationSchema = yup.object({
    name: yup.string().trim().min(3).max(100).required(),
    price: yup.number().min(0).required(),
});

try {
    let validData = await validationSchema.validate(data,
        { abortEarly: false });
    // TODO: Process valid data
}
catch (err) {
    res.status(400).json({ errors: err.errors });
}
```

- **name**: It should be a string, with leading and trailing whitespace trimmed, having a minimum length of 3 characters, a maximum length of 100 characters, and it is required (cannot be empty or missing).
- **price**: It should be a number, with a minimum value of 0, and it is required.

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Validate Input

- Validate input data after the schema is defined

```
let data = req.body;
// Validate request body
let validationSchema = yup.object({
    name: yup.string().trim().min(3).max(100).required(),
    price: yup.number().min(0).required(),
});

try {
    let validData = await validationSchema.validate(data,
        { abortEarly: false });
    // TODO: Process valid data
}
catch (err) {
    res.status(400).json({ errors: err.errors });
}
```

- The **validate()** method is used to validate data against the schema and returns the valid data and any applied transformations.
- The **abortEarly** option is set to false, which means that validation will continue even if there are multiple validation errors.
- If there are validation **errors**, the code enters the catch block. It sends a response with a status code of 400 (Bad Request) and includes the error messages.

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Yup Functions

Data Transformation

- **Common**: required(), oneOf(array), default(value)
- **String**: string(), trim(), lowercase(), uppercase(), length(limit), min(limit), max(limit), matches(regex), email(), url()
- **Number**: number(), min(limit), max(limit), integer()
- **Date**: date(), min(limit), max(limit)
- **Boolean**: boolean()

JSON date value example

- **Date Only**: “2020-11-20” for 20 Nov 2020
- **Date Time**: “2020-11-20T16:50:00Z” for 20 Nov 2020 16:50:00

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Input Sanitization

Input sanitization is the process of cleaning input data to ensure it is safe and reliable before it is processed further.

- **Security:** Prevent security vulnerabilities and protect against attacks such as SQL injection, cross-site scripting (XSS), and command injection.
- **Data Integrity:** Ensure the data meets the required quality standards for processing and storage. This can include removing whitespace, converting text to lowercase, or reformatting data for consistency.

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Summary

1. Describe Sequelize ORM.
2. Understand Sequelize CRUD Operations.
3. Describe Sequelize Associations.
4. Implement Input Validation using Yup.

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Credits

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References:

- Coding stickers created by kerismaker - Flaticon
  https://www.flaticon.com/stickers-pack/web-development-195

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